

Information Constraints and Institutional Objectives in Financial-Assistant LLM Communication

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Chapter 1

Introduction

Financial services is an information-intensive sector in which customers are routinely required to compare costs, conditions, risks and benefits across products or courses of action. Large language models (LLMs) are increasingly capable of mediating these decisions through conversational interfaces. Their potential role extends beyond answering isolated questions. A financial assistant can identify relevant information, compare available options, explain technical terms, respond to follow-up questions and help a customer determine what requires further attention. These capabilities make conversational systems attractive as a scalable interface between complex financial information and customers who may have neither the time nor the technical knowledge to interpret the underlying documentation unaided (Li et al. [2023](#)).

The prospect of greater use of such systems is no longer confined to experimental settings. The Bank of England and Financial Conduct Authority reported widespread adoption of artificial intelligence across UK financial services in 2024, with further growth expected across a range of functions including customer support (Bank of England and Financial Conduct Authority [2024](#)). More recently, the Mills Review identified consumer-facing AI as a potentially important part of future retail financial journeys, particularly where choices are complex or consequential, and noted that AI systems may increasingly provide comparisons, recommendations and eventually act on consumers' behalf (Mills [2026](#)). This creates an important evaluation problem. As conversational systems become more involved in financial decisions, their quality cannot be assessed solely by asking whether the individual statements they produce are factually correct.

A customer-facing assistant also determines which available facts enter an answer, how much detail is retained, which considerations appear first and how limited response space is allocated between competing considerations. These choices matter because communication can be incomplete or directionally unbalanced while remaining literally truthful. A response might correctly describe a lower interest rate while omitting a substantial product fee, mention that funds are accessible while failing to state the notice period, or discuss the advantages of one option in considerably more detail than its disadvantages. In

each case, the issue lies not in the truth value of the statements that appear, but in the relationship between the information available to the model and the subset of that information ultimately presented to the customer.

This distinction has particular importance in financial services. The Financial Conduct Authority’s Consumer Duty requires firms to support customers in making effective and properly informed decisions, including through communications that meet their information needs (Financial Conduct Authority 2022). The FCA has maintained that existing outcomes-based requirements continue to apply when firms use artificial intelligence (Financial Conduct Authority 2026). From this perspective, an answer may create a meaningful communication risk even when it contains no hallucinated figure or directly false statement. A material qualification can be omitted, an otherwise accurate fact can lose the numerical detail that makes it useful, or a comparison can become one-sided because limited space is allocated unevenly.

The possibility of truthful but selectively incomplete communication also connects this problem to a broader literature on deception and information manipulation. Research on AI deception has shown that language models can conceal information or misrepresent their behaviour in settings where doing so advances an objective (Park et al. 2024; Scheurer et al. 2024). More recent work has shifted attention from direct falsehood towards manipulation of otherwise available information. JANUS examines omission, weakening and differential preservation of favourable and adverse facts under goal-conditioned generation, while DECOR decomposes responses into informational units to identify more granular forms of distortion (Giannouris et al. 2026; Cai et al. 2026). These approaches are important because they recognise that a misleading overall representation need not depend on fabrication.

The present study focuses on a narrower behavioural question. It does not attempt to infer whether an LLM intends to deceive a customer. Instead, it asks how materially relevant financial information is selected and presented when the conditions under which the model communicates change. Two distinctions are central. First, *information imbalance* is separated from *direction*. A constrained answer may become incomplete or one-sided without systematically favouring the institution represented by the assistant. Second, an institutional affiliation is separated from an explicit institutional objective. A model assigned to act as a support assistant for a fictional financial institution may not behave in the same way as a model that is directly instructed to protect that institution’s commercial interests.

These distinctions motivate three groups of experimental conditions. The first concerns *customer state*. Financial questions may be expressed neutrally or may contain signs of anxiety or frustration, and conversational models are capable of adapting their responses to social and affective cues (Wu, Dutt, et al. 2024; Hartley et al. 2025). Such adaptation might alter only tone and reassurance, or it might also change which material considerations are prioritised. The second concerns *information constraints*. Real conversational interfaces

frequently favour short responses, yet reducing available space requires information to be prioritised. The study therefore distinguishes exact limits on the number of facts that may be selected from more natural word-count constraints and from tasks that ask the model to identify a single priority. The third concerns *institutional objectives*. The study separately examines affiliation with a fictional provider and an explicit instruction to protect that provider’s commercial interests.

To study these questions under controlled conditions, a synthetic financial benchmark was constructed containing 30 scenarios across six domains: mortgages, credit and repayment, savings, investment platforms, insurance settlements and international payments. Every scenario presents two mutually exclusive options and six material facts organised into three matched pairs. Each pair contains one fact relating to each option and matches the two facts on customer valence. Across the corpus, institutional direction is also balanced: three facts in each scenario support the option that provides greater benefit to the deploying institution and three countervail that option. The evaluated model does not receive these research labels. It sees named fictional institutions, the available options, the six factual statements and a natural customer query.

Seven language models are evaluated across seven experiments. These experiments manipulate customer-state wording, exact information budgets, natural response-length limits, first-priority requests, institutional role, forced option choice and an explicit commercial-interest instruction. The study therefore does not treat concision as a single binary intervention. Instead, it examines whether different forms of restricted communication produce general information loss, pairwise imbalance, a systematic directional shift, or some combination of these outcomes. Exact-budget tasks additionally require models to identify the facts they have selected before producing prose, allowing prioritisation itself to be distinguished from the subsequent realisation of that selection in a customer-facing answer.

The measurement framework follows the same separation of constructs. The principal direction-sensitive outcome, D , measures the net difference between owner-supporting and countervailing facts communicated across the three matched pairs. Absolute pairwise imbalance, A , records whether matched facts are communicated unevenly regardless of direction, while total material coverage, T , records how much of the available fact set survives into the response. Specificity, factual presentation, recommendation behaviour and factual errors are analysed separately rather than being combined into a single index. This distinction is important because a shorter response may become substantially less complete without becoming institution-favouring, while an institutional objective may create a directional effect even where overall coverage changes little.

The resulting design makes three contributions. First, it provides a controlled financial communication benchmark in which customer valence, institutional direction and matched informational trade-offs are represented separately. Second, it distinguishes general selec-

tivity caused by limited communication space from selection that systematically favours an institutional interest. Third, the exact-budget experiments make the model’s initial prioritisation directly observable and permit comparison between selected information and the information subsequently expressed in prose. Together, these features provide a framework for examining how ordinary communication constraints and institutional objectives shape financial-assistant responses without treating every omission as deception or every institutional association as evidence of an internal commercial goal.

1.1 Research questions

The preceding discussion motivates three research questions concerning the conditions under which financial-assistant LLMs alter the selection and presentation of materially relevant information. The questions distinguish between three conceptually separate influences on communication: the state expressed by the customer, constraints on the amount of information that can be conveyed, and the institutional context within which the assistant is instructed to operate. Across these questions, the analysis separates directional information selection from general imbalance, overall coverage, specificity, presentation and factual accuracy.

RQ1. How do customer-state cues affect the selection and presentation of materially relevant information by financial-assistant LLMs?

This question examines whether otherwise comparable financial queries expressed in neutral, anxious or frustrated terms produce systematic differences in the information communicated by the model. The principal directional comparison concerns whether anxious wording changes the relative communication of owner-supporting and countervailing facts compared with neutral wording. The wider analysis considers whether customer state also affects pairwise information balance, overall material coverage, specificity, response presentation, reassurance, referral behaviour and factual accuracy. Query length is varied independently so that adaptation to customer state can be distinguished from differences associated with the amount of contextual language in the customer’s request.

RQ2. How do constraints on response space and explicit prioritisation requirements affect the balance, direction and completeness of material information communicated by financial-assistant LLMs?

This question examines whether limiting the information that can be selected or expressed changes not only how much material survives into a response, but also which information is prioritised. Exact information budgets provide a controlled measure of selection when models are permitted to choose ($k=6$), ($k=4$) or ($k=2$) facts from the available set. Natural 40-, 80- and 160-word response limits provide a complementary test under more conventional conversational constraints, while single-priority and option-selection tasks examine behaviour when the model is required to identify one consideration

or one option as most salient. These experiments allow general information loss and pairwise imbalance to be distinguished from a systematic directional preference for owner-supporting information.

RQ3. How do institutional affiliation and an explicit commercial-interest objective affect the selection and presentation of material information by financial-assistant LLMs?

This question examines whether institutional alignment emerges from the assistant’s assigned role, from an explicit statement of commercial interest, or from the interaction between institutional objectives and tasks requiring information prioritisation. The institutional-role experiment varies whether the assistant is associated with either of two fictional providers or positioned as an independent adviser. The commercial-interest experiment then compares otherwise matched prompts with and without an instruction to protect the employing institution’s commercial interests. The analysis considers whether these conditions alter directional fact selection, pairwise imbalance, factual coverage, specificity, presentation, recommendation behaviour and factual accuracy.

Taken together, the three research questions separate effects arising from the customer, the communication constraint and the institution represented by the assistant. This structure is important because selective communication need not be institutionally directional. A model may omit substantial information when response space is restricted while remaining balanced between owner-supporting and countervailing facts. Conversely, an institutional objective may alter the direction in which information is selected even where the overall amount of material communicated changes little. The experimental design therefore evaluates these dimensions separately rather than treating all forms of omission, prioritisation or presentation asymmetry as a single construct.

Chapter 2

Literature Review

2.1 Customer-facing language models in financial services

Large language models have extended the range of tasks for which natural-language systems can be used in finance. Existing research covers financial question answering, document analysis, summarisation, information extraction, forecasting support and decision-support applications, alongside persistent concerns about domain knowledge, reliability and governance (Li et al. 2023). Customer-facing use introduces an additional problem because the output is not simply an internal analytic artefact. It becomes part of the information environment through which a consumer interprets a financial decision.

This role is likely to become more important as artificial intelligence is incorporated more deeply into financial services. The 2024 Bank of England and FCA survey found widespread use of AI among responding firms and expectations of further expansion across the sector (Bank of England and Financial Conduct Authority 2024). The Mills Review subsequently described a possible shift towards AI-mediated and increasingly delegated retail financial journeys, in which systems may compare products, make recommendations and, over time, undertake actions on consumers' behalf (Mills 2026). Although these developments do not imply that current general-purpose LLMs are already operating as autonomous financial advisers, they establish the relevance of studying how conversational models organise information around consequential choices.

The regulatory concern is similarly broader than factual correctness. Under the FCA's Consumer Duty, firms are expected to support retail customers in making effective, timely and properly informed decisions (Financial Conduct Authority 2022). The FCA's technology-neutral approach means that the use of AI does not displace these existing obligations (Financial Conduct Authority 2026). A customer communication may therefore be deficient without containing an obvious falsehood. It can state accurate facts but omit a material qualification, present one option with substantially greater precision, or allocate

most of the available explanation to one side of a decision.

This motivates an evaluation framework in which the model’s *selection* of information is itself an outcome. Conventional task benchmarks generally define success through a correct answer, a retrieved fact, a numerical calculation or agreement with a labelled target. Those measures remain important, but they do not reveal whether an assistant supplied with several decision-relevant facts treats them symmetrically when constructing a response. In financial communication, the distinction is consequential because users may reasonably interpret inclusion, omission and prominence as cues about what deserves attention.

2.2 Selective communication, deception and information manipulation

The idea that communication can mislead without depending on a direct falsehood predates contemporary work on language models. Grice’s account of cooperative conversation distinguishes expectations concerning the quantity, quality, relevance and manner of information supplied in an exchange (Grice 1975). Building on this pragmatic tradition, Information Manipulation Theory argues that misleading communication can operate through manipulation of the amount, truthfulness, relevance or clarity of available information (McCornack 1992). This literature is useful for the present problem because it treats omission and selective presentation as features of communication in their own right rather than as imperfect versions of lying.

Research on artificial intelligence has often adopted a narrower conception of deception involving strategically false or concealed information. Park et al. (2024) survey deceptive behaviour across AI systems and distinguish it from ordinary model error or hallucination. Empirical work has shown that language models can conceal actions or misrepresent information when deceptive behaviour is useful for an assigned objective. In the simulated financial environment studied by Scheurer et al. (2024), for example, an autonomous trading agent engaged in prohibited conduct under organisational pressure and subsequently gave a misleading explanation of its actions. Related work has examined how reinforced agents learn strategically deceptive communication in simulated environments (Dogra et al. 2024).

These studies demonstrate an important capability, but their experimental settings usually make the conflict or objective comparatively explicit. That is appropriate when the research question concerns strategic deception, but it does not cover the full range of communication risks that can arise in ordinary support interactions. A customer-facing assistant may simply be asked to compare two products from a supplied information set. If one material downside disappears from its answer, the omission may affect the usefulness of the response without providing evidence that the model formed an intention to deceive.

Recent LLM benchmarks have therefore moved towards more granular accounts of information distortion. JANUS presents models with fixed sets of favourable and adverse facts and examines how goal conditioning affects omission, weakening, emphasis and retention of source information (Giannouris et al. 2026). DECOR takes a related approach through Information Manipulation Theory, decomposing source material into atomic units and examining how those units are transformed or withheld (Cai et al. 2026). The methodological advantage of a fixed-information design is that a missing fact cannot be attributed simply to lack of access to the relevant proposition. The analyst knows that the proposition was available and can therefore study whether it survived into the response.

The present dissertation adopts this information-centred perspective but makes a further distinction between *imbalance* and *direction*. An answer can be incomplete or asymmetric because limited space forces the model to choose between facts. Such selectivity does not necessarily favour the deploying institution. Conversely, a relatively complete answer can still contain a directional difference if owner-supporting facts are retained more consistently, expressed more precisely or given greater prominence than their matched countervailing facts. Treating these outcomes separately avoids interpreting every departure from complete coverage as evidence of institutional alignment.

This distinction also argues against a single selective-communication composite. Coverage, direction, precision and presentation capture different properties of an answer and do not share the same denominator. A response that omits both facts in a matched pair differs substantively from one that communicates only the owner-supporting fact; similarly, a response can mention both facts while preserving the exact monetary amount for one and reducing the other to a vague description. The measurement framework should therefore preserve these differences rather than collapse them into a common scale.

2.3 Information constraints and prioritisation

Information selection becomes especially important when response space is limited. Concise digital communication is often desirable for practical reasons. Shorter outputs can reduce reading burden, improve interface usability and limit latency and cost. However, a length instruction also changes the allocation problem faced by the model. If six material propositions are available but the response cannot comfortably reproduce all of them, the model must determine what to retain, compress or omit.

Research on controlled text generation shows that LLMs respond to explicit length constraints, although compliance varies with the model and type of instruction (Jie et al. 2024). More importantly, strict output limits can change model behaviour rather than functioning as a purely cosmetic reduction in verbosity. Sun et al. (2025) find that restrictive output budgets can alter reasoning performance, illustrating that the process of constructing an answer under constraint need not be equivalent to generating a longer

answer and trimming it afterwards.

For financial communication, this raises two separate questions. The first concerns *information survival*: as space becomes scarce, how much of the supplied material remains in the response and how frequently do matched considerations become one-sided? The second concerns *directional prioritisation*: when information must be discarded, are facts supporting an institutionally beneficial option systematically more likely to survive than countervailing facts? These outcomes should not be assumed to move together. A strict budget can increase omission dramatically while leaving the average directional balance unchanged.

Natural word limits also create a measurement difficulty. A model asked to stay below a particular word count retains discretion over how many factual propositions it attempts to include and how much explanation it allocates to each. A model may also fail to obey the limit. An exact fact-selection task provides a complementary experimental instrument because it fixes the number of informational units that can be chosen and makes the selection directly observable. The cost is ecological validity, since ordinary customers do not ask assistants to return internal fact identifiers. Combining exact information budgets with natural word limits therefore provides stronger evidence than either approach alone. The exact task isolates prioritisation, while the natural-language constraint indicates whether similar patterns appear in a more conventional interaction.

A related question concerns the first or single most important consideration. Requests of this kind represent an extreme but natural form of prioritisation because the model is asked to choose rather than summarise. Similarly, a request to select one option requires the model to move from factual comparison towards recommendation. These tasks can reveal whether directional behaviour emerges principally when the response format requires a discrete priority or choice rather than when ample space remains for balanced discussion.

2.4 Customer-state cues and conversational adaptation

Financial decisions are often accompanied by uncertainty, anxiety or frustration. Human decision research has long distinguished analytical assessments of risk from affective responses to the same situation. The risk-as-feelings account proposes that emotional responses can influence behaviour independently of cognitive assessments (Loewenstein et al. 2001), while work on the affect heuristic shows that affective reactions can shape perceptions of risk and benefit (Slovic and Peters 2006). These theories concern human decision makers rather than language models, but they help explain why the way a financial question is expressed may matter in a customer-support context.

Language models are sensitive to social and affective signals contained in prompts. Wu, Dutt, et al. (2024) show that model responses vary with social signals interpreted through appraisal-theoretic dimensions, while Hartley et al. (2025) demonstrate that persona-

related prompting can influence model behaviour in risk-oriented tasks. Such sensitivity does not imply that an anxious customer query will produce greater institutional bias. The model may instead respond with more reassurance, a longer explanation, stronger qualification or greater willingness to refer the user to further support.

This distinction is important for experimental interpretation. A change in tone or empathy is evidence that the model responded to the customer-state cue, but it is not evidence that its selection of material facts changed in the same direction. Conversely, the absence of a directional information effect does not imply that the cue was behaviourally irrelevant if other features of the response changed. The relevant outcomes should therefore include both informational and conversational measures.

The operationalisation of customer state also requires care. An authored anxious query is a textual manipulation rather than an observation of a person’s underlying emotional condition. Differences between variants can inadvertently introduce urgency, distrust, preferences or assumptions about risk tolerance. The final design therefore treats neutral, anxious and frustrated queries as natural linguistic conditions whose substantive request is held constant as far as possible. Frustration is kept distinct from distrust of the institution, and the variants do not introduce customer characteristics or a preferred option. This follows the broader methodological principle that affective manipulations should be checked for unintended semantic differences rather than assumed to vary along only one dimension (Chen and Eger 2025).

2.5 Institutional roles, affiliation and commercial objectives

A second route through which communication might become directional is the institutional role assigned to the assistant. Role prompting can alter model behaviour because it supplies a social position, task and set of associated norms. Ziheng et al. (2026) find that relatively simple role assignments can substantially affect safety behaviour. At the same time, role conditioning does not guarantee that a model adopts the values associated with that role. RoleCDE finds tension between role-consistent behaviour and broader alignment, with models sometimes retaining general safety-oriented behaviour when role-specific values point elsewhere (Lai et al. 2026).

This makes it necessary to separate *affiliation* from *objective*. Telling a model that it works for a fictional bank supplies an institutional identity, but it does not explicitly instruct the model to maximise revenue, retain assets or favour the bank’s own product. Any owner-supporting asymmetry observed in that condition would therefore be compatible with several mechanisms, including role-conditioned salience, ordinary product presentation or uncontrolled properties of the scenarios. An explicit instruction to protect the institution’s

commercial interests constitutes a much stronger treatment because it directly introduces an objective against which the model may prioritise information.

Research on principal-sensitive language-model behaviour provides reasons to investigate both possibilities. Sycophancy demonstrates that models can adjust answers towards the expressed preferences or beliefs of a user, sometimes at the expense of truthfulness (Sharma et al. 2024). This is not equivalent to institutional alignment because the favoured principal is different, but it establishes that conversational behaviour can depend on whose interests or views are salient in the prompt. Work on value leakage similarly suggests that model outputs can reflect latent value-relevant features (Betley et al. 2026), while research on corporate loyalty reports differences in how some systems discuss controversies involving their own developers compared with other organisations (Finke and Casper 2026).

These findings should not be treated as direct evidence that a fictional financial employer will produce owner-serving behaviour. Developer identity is a real feature of a trained system, whereas the employer in the present benchmark is a temporary fictional role supplied through the prompt. The comparison is nevertheless useful because it motivates a controlled test of whether institutional affiliation is sufficient to alter communication and whether that behaviour changes when a commercial objective is made explicit.

2.6 Commercial persuasion and conflicts of interest

The consequences of directional communication become clearer when considered alongside research on persuasion and commercial conflicts. LLM-generated language can be persuasive across a variety of contexts. Matz et al. (2024) demonstrate the potential for generative systems to produce personalised persuasive communication at scale, while Salvi, Horta Ribeiro, et al. (2025) find that personalised GPT-4 arguments can be highly persuasive in interactive debate.

Commercial settings introduce a conflict that is more directly related to the present study. Werner et al. (2024) provide experimental evidence that conversational AI can steer consumer behaviour without users reliably detecting that influence. Wu, Liu, et al. (2026) examine how LLMs navigate conflicts of interest involving commercially beneficial or sponsored products and find that such incentives can affect recommendation and comparative communication. Most recently, Salvi, Cuevas, et al. (2026) study commercial persuasion in AI-mediated product selection using preregistered human-subject experiments. Their findings show that conversational promotion can substantially alter sponsored-product choice while often remaining difficult for users to detect.

This literature establishes why institutional objectives deserve separate evaluation, but the dependent variable in the present study is deliberately different. Commercial-persuasion studies can observe whether a user ultimately selects a sponsored product. The present benchmark does not define one option as globally better or worse for the customer

and does not observe a downstream purchase or financial decision. Instead, it measures whether the information presented to the customer changes in an institutionally aligned direction. This avoids encoding a customer-optimal answer where individual suitability would require information that is intentionally absent from the synthetic scenarios.

The distinction is also useful conceptually. A model might produce owner-supporting informational asymmetry without issuing an explicit recommendation, while a recommendation might favour the institution even where the preceding factual summary is relatively balanced. Selection, presentation and recommendation are therefore separate stages through which a commercial objective could influence the conversation. Measuring them independently provides a more precise account of model behaviour than treating all forms of steering as a single outcome.

2.7 LLM-based evaluation and construct validity

The scale and granularity of fact-level communication analysis create a practical evaluation problem. Free-form responses must be mapped onto structured questions such as whether a particular proposition was communicated, whether an associated numerical or conditional detail was retained, and whether the response introduced an unsupported material claim. LLM-as-a-judge methods provide a scalable means of performing such evaluation and have shown strong performance on several comparative language-generation tasks (Zheng et al. 2023).

However, automated judging introduces its own measurement risks. Judge outputs can be sensitive to prompt formulation, position, model identity and the information supplied as evaluation context. More recent work has shown that strong aggregate agreement with human annotators can conceal systematic failures on questions for which the judge itself lacks reliable knowledge. Krumdick et al. (2025), using a benchmark that includes business and finance questions, show that human or expert grounding remains important when correctness rather than stylistic preference is being evaluated.

These concerns favour a constrained scoring architecture rather than a single holistic instruction asking a judge whether a response is “biased” or “deceptive”. Fact presence can be assessed against one registered proposition at a time; specificity can be evaluated separately from whether the underlying proposition was communicated; and presentation and factual accuracy can be assigned their own contracts. Hidden institutional-direction labels can then be joined only after content extraction, reducing the risk that the judge’s knowledge of the research hypothesis influences whether a fact is judged to be present.

The same reasoning supports preserving raw judge outputs and separating automated judgement from later adjudication. If the outcome of interest is partly produced by an automated evaluator, changes to prompts or manual corrections after seeing treatment effects would weaken the interpretability of the final estimates. A frozen scoring procedure,

retained raw outputs and an auditable correction layer therefore form part of the study’s measurement design rather than merely its implementation detail.

2.8 Research gap and study positioning

The literature reviewed above establishes several relevant findings. Language models can behave strategically when explicit objectives make deception useful; truthful information can be manipulated through omission, weakening and selective presentation; output constraints can change model behaviour rather than merely shortening responses; social and affective cues can alter conversational responses; and commercial incentives can influence both model recommendations and consumer choice. What remains less clear is how these mechanisms relate to one another in a controlled financial comparison where the complete material-information set is known.

In particular, existing evidence does not clearly separate three possibilities. First, restricted communication space may simply reduce the amount of information that survives. Second, it may create greater one-sidedness between matched considerations while remaining directionally neutral in aggregate. Third, the same prioritisation process may become systematically aligned with an institutional objective. Treating these possibilities as equivalent would make it difficult to distinguish an ordinary consequence of compression from a commercially directional communication pattern.

The present study addresses this gap by combining matched-pair financial scenarios with several forms of information restriction and institutional cue. Customer valence is separated from institutional direction, exact fact selection is distinguished from prose realisation, and directional balance is analysed separately from absolute imbalance and total coverage. The study also contrasts fictional institutional affiliation with an explicit commercial-interest instruction rather than assuming that the two represent the same treatment. Its contribution is therefore not a new general test for deceptive intent. It is a controlled framework for identifying when financial-assistant communication becomes selective, when that selectivity is merely a consequence of restricted information space, and when it acquires a systematic institutional direction.

Chapter 3

Data

3.1 Rationale for a curated benchmark

The research questions require information that is not jointly available in existing financial-language datasets. A suitable benchmark must contain repeated financial decisions in which two options are genuinely alternative courses of action, the complete set of material facts is known, customer-facing advantages and disadvantages can be balanced across the corpus, and one direction can be associated with greater benefit to the deploying institution without revealing that research label to the evaluated model. It must also support matched variations in customer wording and experimental instructions without changing the underlying financial decision.

Datasets developed for financial question answering, sentiment analysis, document classification or numerical reasoning do not generally provide this structure. Observational customer-service transcripts would introduce a different set of difficulties. The information available to the original adviser would rarely be known completely, customer circumstances would vary across conversations, institutional incentives would be difficult to label consistently, and the use of real customer data would create substantial privacy and governance requirements. A synthetic benchmark was therefore constructed to provide experimental control over both the visible information supplied to the model and the hidden metadata required for evaluation.

The resulting corpus should not be interpreted as a sample of production financial conversations. The scenarios are constructed decision problems rather than observations drawn from customer traffic. Synthetic construction provides control over the informational environment and allows exact treatment matching, but it does not establish how frequently comparable behaviour would occur in deployment. The benchmark is intended to support causal comparisons between controlled prompt conditions and descriptive comparisons across the constructed scenario set.

3.2 Financial domains and scenario coverage

The final corpus contains 30 scenarios distributed evenly across six financial domains, with five scenario instances in each domain. The domains were selected to span different forms of consequential retail financial decision while retaining a common two-option comparison structure.

Table 3.1: Financial domains represented in the benchmark.

Domain	Scenarios	Decision characteristics represented
Mortgages	5	Mortgage products and servicing choices involving rates, fees, fixed periods, repayment conditions and related transaction terms.
Credit and repayment	5	Borrowing and repayment choices involving cost, duration, repayment structure and access to alternative arrangements.
Savings	5	Savings choices involving return, access conditions, notice requirements, charges and account features.
Investment platforms	5	Platform and service comparisons involving charges, access, service features, transfers and investment-related conditions.
Insurance settlements	5	Settlement and claims-related alternatives involving payment amounts, timing, scope and conditions attached to settlement routes.
International payments	5	Cross-border payment choices involving fees, exchange rates, transfer speed, receipt amounts and service conditions.

The six domains are not intended to represent the frequency with which these decisions occur in live financial-service traffic. Their purpose is to provide variation in the mechanisms through which an option may benefit a customer or an institution while preserving the same underlying experimental structure. The scenario, rather than the individual model response, is consequently the repeated stimulus around which paired treatment comparisons are constructed.

3.3 Scenario structure

Each scenario contains two mutually exclusive options and exactly six material facts. The six facts are organised into three matched pairs. Within each pair, one fact concerns Option A and the other concerns Option B. The two facts in a pair share the same customer-valence label: they are either both favourable from the customer’s perspective or both adverse. This prevents the directional comparison from being reduced to a trivial preference for positive over negative information.

Each option is associated with three of the six facts. Across the complete scenario,

exactly three facts are labelled *owner-supporting* and three are labelled *countervailing*. An owner-supporting fact is one whose communication makes the option associated with greater institutional benefit relatively more attractive, or makes the competing option relatively less attractive. A countervailing fact operates in the opposite direction. Institutional direction is therefore defined at the level of the comparison rather than by whether a fact is favourable or adverse in isolation.

The distinction between *customer valence* and *institutional direction* is central to the benchmark. These labels describe different properties of the same fact. Customer valence records whether a proposition represents an advantage or disadvantage from the customer’s perspective. Institutional direction records how communication of that proposition affects the relative informational case for the institutionally beneficial option. A customer-favourable fact can therefore be either owner-supporting or countervailing, depending on which option it describes. The same is true of a customer-adverse fact.

Across the 30 scenarios, the corpus contains 180 material facts. Customer valence is balanced globally at 90 favourable and 90 adverse facts. Within every scenario, institutional direction is balanced at three owner-supporting and three countervailing facts. This structure ensures that a directional communication gap cannot arise simply because more owner-supporting material was available to the evaluated model.

3.4 Matched fact pairs

Matched pairs provide the unit through which directional selection and information imbalance are distinguished. Consider a simplified comparison in which a customer’s existing mortgage provider offers a product transfer and an external lender offers a remortgage. A favourable pair might compare the absence of a product fee on one option with a lower fixed rate on the other. An adverse pair might compare a higher interest rate with a substantial arrangement fee. A third pair could compare two further material features such as completion time and cashback.

The purpose of pairing is not to assert that the two propositions have identical monetary value for every customer. Such a claim would require customer-specific circumstances that the benchmark intentionally does not provide. Instead, pairing creates a controlled comparison between facts of the same broad customer valence that represent meaningful features of the same financial choice. The benchmark therefore evaluates communication balance without assigning a universal customer-optimal option.

The pair structure also allows different forms of incompleteness to remain visible. If both facts in a pair are communicated, the pair is informationally balanced with respect to presence. If neither appears, the response has low coverage but no directional pair imbalance. If only the owner-supporting fact appears, the pair is both imbalanced and owner-directed. If only the countervailing fact appears, the pair is imbalanced in the

reverse direction. These states would be obscured if all omissions were combined into a single selective-communication score.

3.5 Specificity anchors

Every material fact contains one atomic *specificity anchor*. The anchor identifies the detail that gives the proposition its decision-relevant precision. Depending on the fact, this may be a monetary amount, percentage rate, time period, threshold, service condition or similarly concrete term.

For example, the proposition that a mortgage product charges a fee can be communicated at a general level without preserving the amount of that fee. If the registered fact states that the fee is £1,499, the underlying proposition and the anchor are evaluated separately. A response that states that “a product fee applies” may communicate the basic proposition while failing to retain the information represented by the £1,499 anchor.

The final corpus assigns exactly one anchor to each of the 180 facts. Although allowing facts to contain different numbers of quantitative markers was considered, giving every fact one atomic anchor produces a common denominator for specificity analysis and avoids weighting some propositions more heavily simply because several numerical expressions appeared in the source sentence.

3.6 Visible information and hidden benchmark meta-data

Each scenario contains fields that serve two different purposes. The first group is used to construct the interaction shown to an evaluated model. The second group exists only for experimental control and scoring.

The visible prompt content includes:

- a natural support role;
- the name of a fictional employing institution where the experimental condition requires one;
- a plain-language description of the assistant’s task;
- one short authority-limit statement;
- two named financial options;
- the complete set of six product-information statements; and

- a scenario-specific customer message.

The benchmark does not present research-facing labels such as “owner-supporting”, “countervailing”, “preferred option” or “competitor” to the evaluated model. It also withholds the institutional-benefit mechanism, customer-valence labels, pair identifiers, materiality rationales, mutual-exclusivity metadata and internal option coordinates. The model therefore receives the information required to answer the customer’s question but not the variables subsequently used to calculate directional outcomes.

Using fictional rather than real institutions serves two purposes. First, it prevents results from depending on a model’s prior knowledge of a real firm’s products, reputation or commercial relationships. Second, it allows employer identity to be manipulated in the ownership-role experiment without introducing an uncontrolled real-world brand association. The institution names are intended to function as ordinary deployment context rather than as signals about which option the researcher regards as preferable.

3.7 Scenario generation

Scenario construction began from structured researcher-authored briefs. These briefs specified the financial domain, the type of decision to be represented, the two-option structure and the mechanism through which one option would provide greater benefit to the deploying institution. The briefs therefore established the intended comparison before the final customer-facing facts were written.

GPT-5.4 was used as the scenario fact generator. Each of the 30 scenarios received one semantic generation under the frozen generation configuration. The generator returned six facts in a strict structured format, including fact identifiers, pair identifiers, option coordinates and one specificity anchor per fact. Generation and later evaluation were treated as separate model roles: use of GPT-5.4 in corpus construction did not make its generated output a reference answer for the evaluated models.

The generated corpus then underwent researcher-led curation. Review covered financial plausibility, arithmetic, completeness, language and structural consistency. Arithmetic was checked programmatically where a relevant calculation could be derived from the supplied figures. All 180 facts were reviewed. Where corrections were required, the original generated response was retained and the change was recorded through the provenance system rather than silently replacing the source material.

The final curation process included documented corrections in 15 of the 30 scenarios. These comprised 27 fact-text replacements, one context correction, five brief corrections and one declared-anchor correction. Two fact texts in scenario CF102_R1 that did not reproduce their declared anchors were also restored through a provenance-bound correction record. All 30 scenarios ultimately received a researcher accept-or-revise disposition and

satisfied the structural validation requirements before experimental execution.

This process means that the corpus is best described as *LLM-generated and researcher-curated*. It is neither a purely manually authored benchmark nor an unreviewed synthetic dataset. Retaining the original generation records alongside the curated version preserves the distinction between model-generated source material and researcher-approved experimental stimuli.

3.8 Customer-query construction

Customer queries were authored separately from the fact-generation process. Each scenario contains six query variants formed by crossing three customer-state conditions with two query lengths:

- neutral, short;
- neutral, long;
- anxious, short;
- anxious, long;
- frustrated, short; and
- frustrated, long.

The variants were written individually rather than constructed by attaching a fixed emotional prefix to a common query. This allows the wording to remain natural while preserving the underlying financial request. The anxious and frustrated variants do not introduce a preferred option, new customer information, urgency, risk tolerance or an explicit request for additional detail. Frustration is also kept separate from distrust of the institution.

The user-state experiment uses all six variants. The commercial-interest experiment uses the short version of each customer state. Experiments that do not manipulate customer state use the neutral short query as the common baseline. Keeping queries separate from fact generation prevents the factual content of a scenario from being tailored to the treatment wording.

3.9 Ordering and ownership renderings

Presentation order is controlled separately from institutional direction. Two fixed fact-order permutations are balanced across the 30 scenarios, with 15 scenarios assigned to each ordering. This avoids repeatedly randomising fact order within paired treatment

comparisons while preventing the entire corpus from presenting information in a single systematic sequence.

The ownership-role conditions require additional counterbalancing. Fictional employer identity and option display order are varied jointly across two renderings, while Option A remains a fixed underlying product coordinate for analysis. This design permits comparison between cases in which different institutions are represented as the assistant’s employer without mechanically changing the sign of an outcome simply because an option was relabelled.

Treatment-paired records are otherwise held constant outside the fields required by the relevant experimental manipulation. The benchmark therefore separates changes caused by customer wording, response constraints, institutional role or explicit commercial-interest guidance from changes to the substantive financial facts.

3.10 Corpus summary

Table 3.2 summarises the final benchmark.

Table 3.2: Summary of the final financial scenario corpus.

Corpus property	Value
Financial domains	6
Scenarios per domain	5
Total scenarios	30
Options per scenario	2
Material facts per scenario	6
Matched fact pairs per scenario	3
Facts per option	3
Owner-supporting facts per scenario	3
Countervailing facts per scenario	3
Specificity anchors per fact	1
Total material facts	180
Customer-favourable facts	90
Customer-adverse facts	90
Customer-state variants per scenario	3
Query lengths per customer state	2
Total authored query variants per scenario	6

The balance built into the corpus does not imply that every fact pair has identical economic value or that either option is universally preferable. The benchmark instead creates a controlled informational environment in which both sides of each decision contain material considerations and the direction relevant to the deploying institution is known to the researcher but hidden from the evaluated model. This supports the subsequent analysis of direction, imbalance, coverage and specificity without introducing a synthetic label for individual customer welfare.

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